

Dual-Clock Comparator (DCC) Module

This chapter describes the dual-clock comparator (DCC) module.

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11.1 Introduction

The primary purpose of a DCC module is to measure the frequency of a clock signal using a second known clock signal as a reference. This capability can be used to ensure the correct frequency range for several different device clock sources, thereby enhancing the system safety metrics.

11.1.1 Main Features

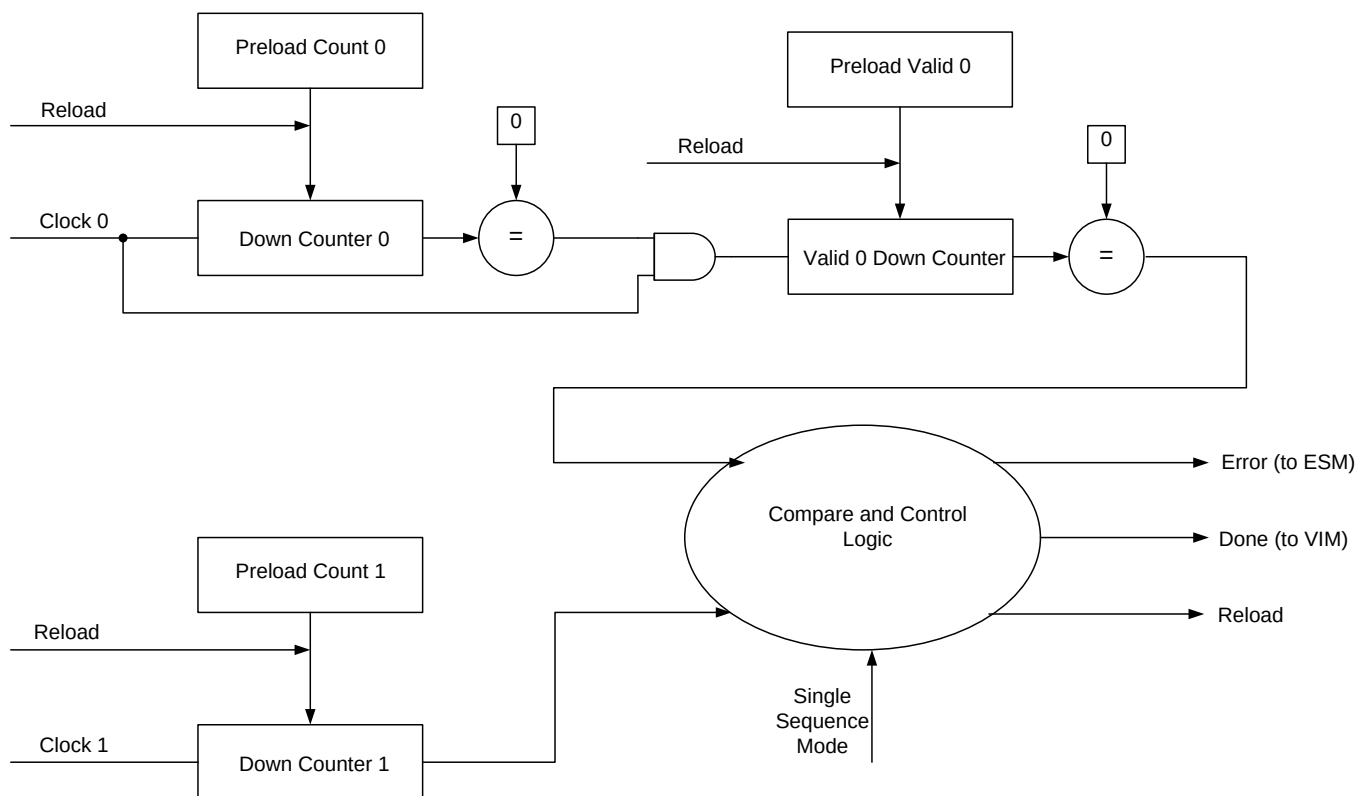
The main features of each of the DCC modules are:

- Allows application to ensure that a fixed ratio is maintained between frequencies of two clock signals
- Supports the definition of a programmable tolerance window in terms of number of reference clock cycles
- Supports continuous monitoring without requiring application intervention
- Also supports a single-sequence mode for spot measurements
- Allows selection of clock source for each of the counters resulting in several specific use cases

11.1.2 Block Diagram

Figure 11-1 illustrates the main concept of the DCC module.

Figure 11-1. DCC Operation



11.2 Module Operation

As shown in [Figure 11-1](#), the DCC contains two counters – counter0 and counter1, which are driven by two signals – clock0 and clock1. The application programs the seed values for both these counters. The application also configures the tolerance window time by configuring the valid counter for clock0.

Counter0 and counter1 both start counting simultaneously once the DCC is enabled. When counter0 counts down to zero, this automatically triggers the count down of the tolerance window counter (valid0).

The DCC module can be used in two different operating modes:

11.2.1 Continuous Monitoring Mode

In this mode, the DCC is used by the application to ensure that two clock signals maintain the correct frequency ratio. Suppose the application wants to ensure that the PLL output signal (clock source # 1) always maintains a fixed frequency relationship with the main oscillator (clock source # 0).

- In this case, the application can use the main oscillator as the clock0 signal (for counter0 and valid0) and the PLL output as the clock1 (for counter1).
- The seed values of counter0, valid0 and counter1 are selected such that if the actual frequencies of clock0 and clock1 are equal to their expected frequencies, then the counter1 will reach zero either at the same time as counter0 or during the count down of the valid0 counter.
- If the counter1 reaches zero during the count down of the valid0 counter, then all the counters (counter0, valid0, counter1) are reloaded with their initial seed values once valid0 has also counted down to zero.
- This sequence of counting down and checking then continues as long as there is no error, or until the DCC module is disabled.
- The counters also all get reloaded if the application resets and restarts the DCC module.

Error Conditions:

An error condition is generated by any one of the following:

1. Counter1 counts down to 0 before Counter0 reaches 0. This means that clock1 is faster than expected, or clock0 is slower than expected. It includes the case when clock0 is stuck at 1 or 0.
2. Counter1 does not reach 0 even when Counter0 and Valid0 have both reached 0. This means that clock1 is slower than expected. It includes the case when clock1 is stuck at 1 or 0.

Any error freezes the counters from counting. An application may then read out the counter values to help determine what caused the error.

11.2.1.1 Error Conditions

While operating in continuous mode, the counters get reloaded with the seed values and continue counting down under the following conditions:

- The module is reset or restarted by the application, OR
- Counter0, Valid 0 and Counter1 all reach 0 without any error

Figure 11-2. Counter Relationship

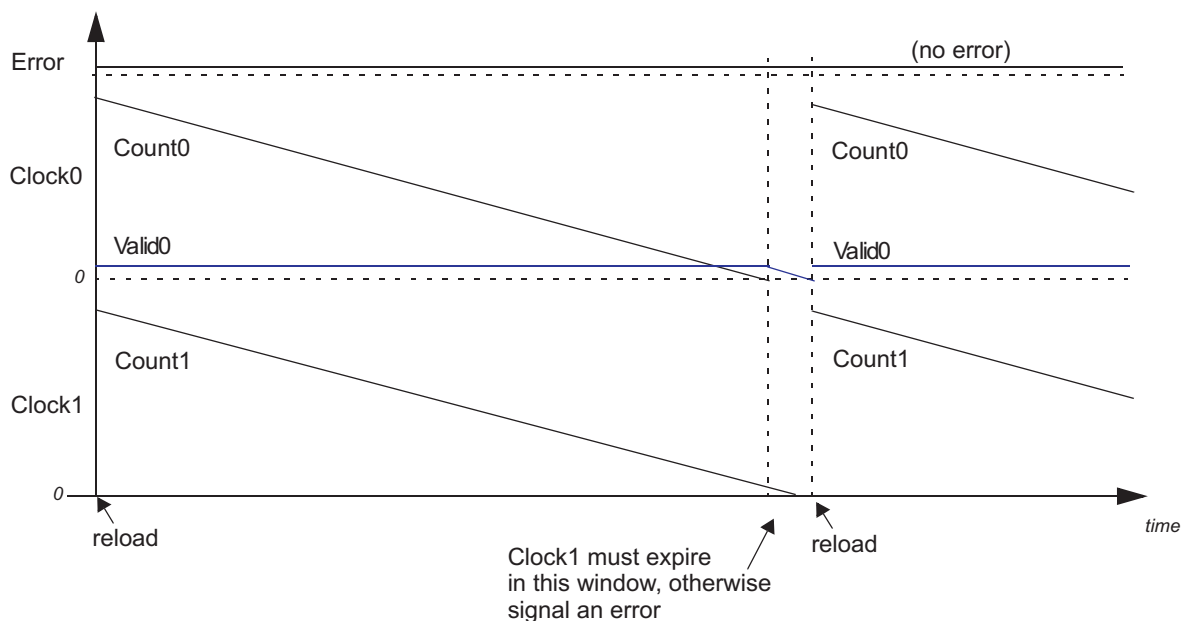


Figure 11-3. Clock1 Slower Than Clock0 - Results in an Error and Stops Counting

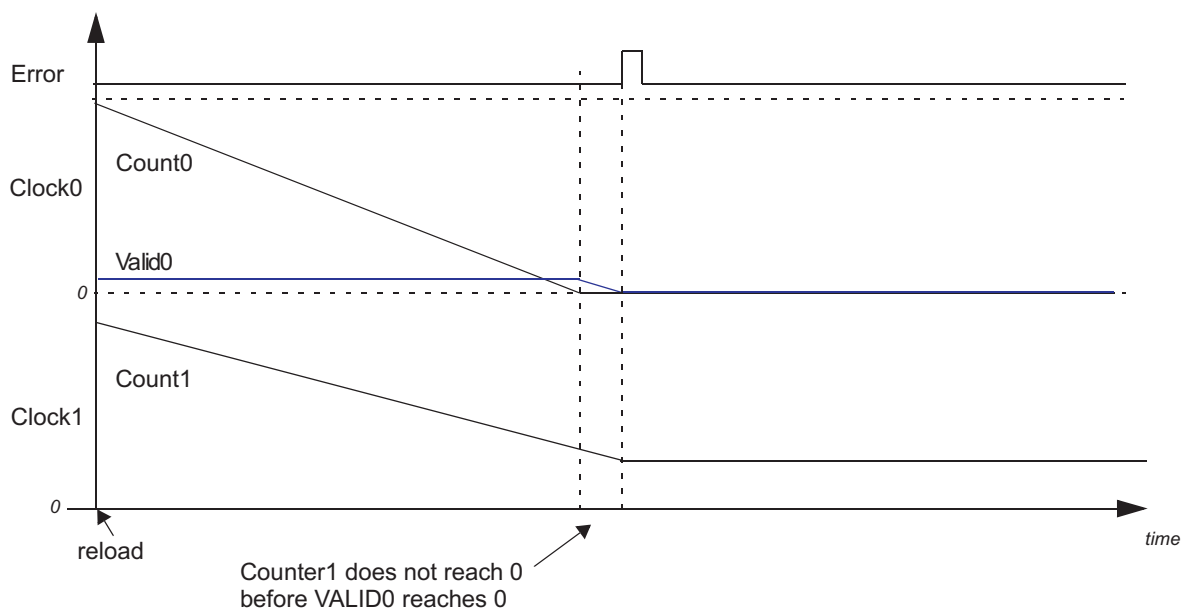


Figure 11-4. Clock1 Faster Than Clock0 - Results in an Error and Stops Counting

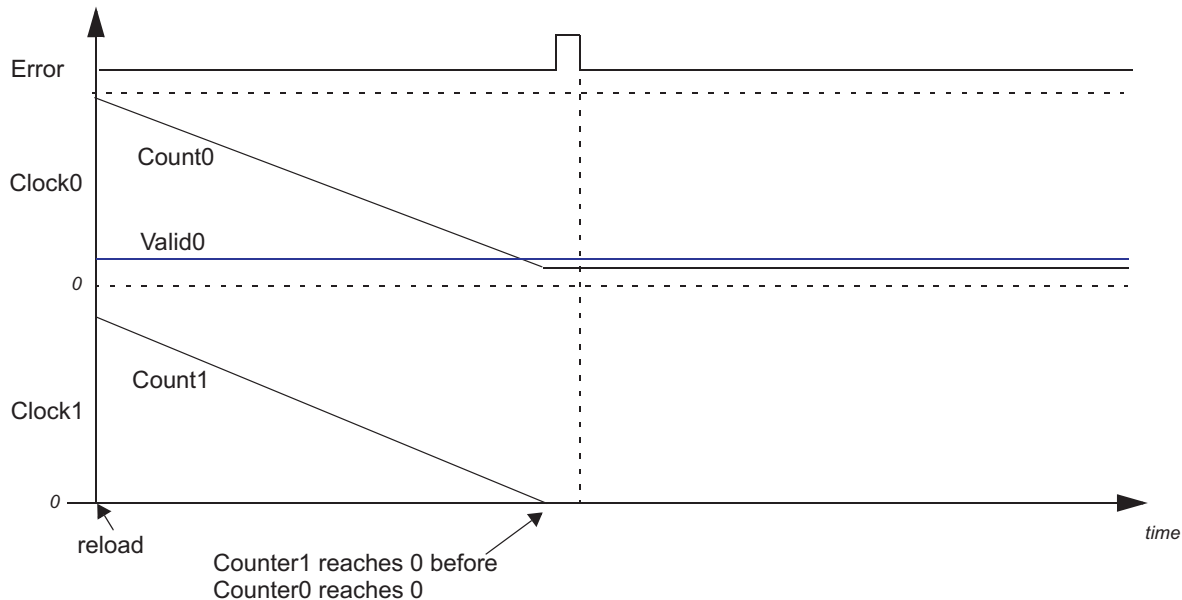


Figure 11-5. Clock1 Not Present - Results in an Error and Stops Counting

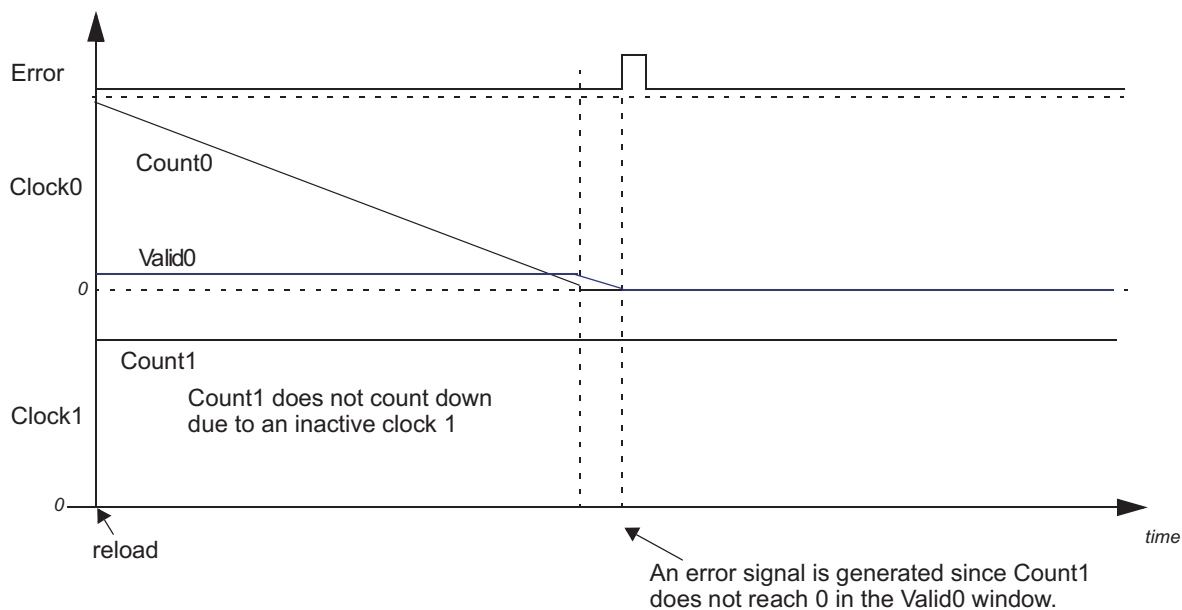
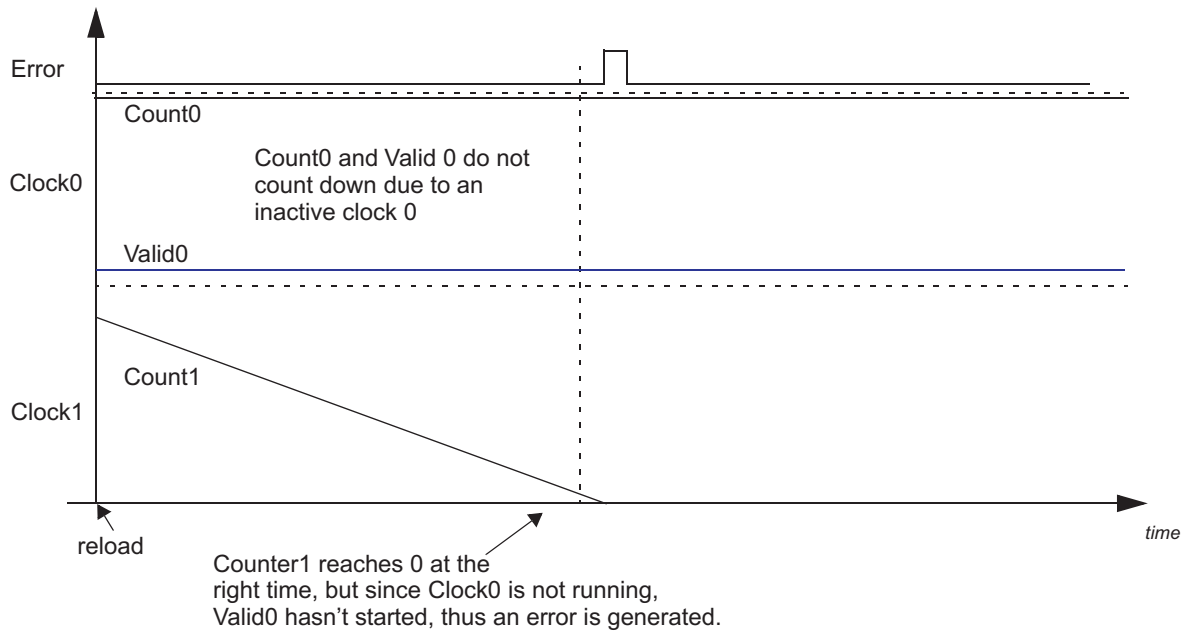


Figure 11-6. Clock0 Not Present - Results in an Error and Stops Counting


11.2.2 Single-Shot Measurement Mode

The DCC module can be programmed to count down one time by enabling the single-shot mode. In this mode, the DCC stops operating when the down counter0 and the valid counter0 reach 0. Alternatively, the DCC can be programmed to stop counting when the down counter1 reaches 0.

At the end of one sequence of counting down in this single-shot mode, the DCC gets disabled automatically, which prevents further counting. This mode is typically used for spot measurements of the frequency of a signal. This frequency could be an unknown for the application before the measurement.

Example Usage of Single-Shot Measurement Mode: Trimming the High-Frequency Low-Power Oscillator

A practical example of the usage of the spot measurement mode is in trimming the HF LPO (clock source # 5) using the main oscillator as a reference. This measurement sequence would proceed as follows:

- The application sets up the seed values for counter0 and valid0 for the duration of the measurement. Suppose the main oscillator frequency is 10MHz and the intended duration of the measurement is 500μs. The application needs to configure a seed value of 5000.
- These 5000 counts need to be divided between the counter0 and the valid0 counters. The minimum value for the valid0 seed is 4, so the application can configure counter0 seed value as 4996 and the valid0 seed value as 4.
- Suppose the HF LPO frequency is truly unknown. In this case the application can choose the maximum allowed seed value for counter1. This increases the probability of counter0 and valid0 counting down while the counter1 has still not fully counted down to zero. The maximum allowed seed value for counter1 is 1048575.
- Once the DCC is enabled, the counters counter0 and counter1 both start counting down from their seed values.
- When counter0 reaches zero, it automatically triggers the valid0 counter.
- When valid0 reaches zero, if counter1 is not zero as well, an ERROR status flag is set and a "DCC error" is sent to the ESM. Counter1 is also frozen so that it stops counting down any further. The application can enable an interrupt to be generated from the ESM whenever this DCC error is indicated. Refer the device datasheet to identify the ESM group and channel where the DCC error is connected.

- The DCC error interrupt service routine can then check the value of counter1 when the error was generated. Suppose that the counter1 now reads 1044575. This means that counter1 has counted 1048575 - 1044575, or 4000 cycles within the 500µs measurement period. This means that the average frequency of the HF LPO over this 500µs period was 4000 cycles / 500µs, or 8MHz.
- The application then needs to clear the ERROR status flag and restart the DCC module so that it is ready for the next spot measurement.

If there is no error generated at the end of the sequence, then the DONE status flag is set and a DONE interrupt is generated. The application must clear the DONE flag before restarting the DCC.

The conditions that cause a DCC error are identical between the continuous monitoring mode and the single-shot measurement mode.

Error Conditions:

An error condition is generated by any one of the following:

1. Counter1 counts down to 0 before Counter0 reaches 0. This means that clock1 is faster than expected, or clock0 is slower than expected. It includes the case when clock0 is stuck at 1 or 0.
2. Counter1 does not reach 0 even when Counter0 and Valid0 have both reached 0. This means that clock1 is slower than expected. It includes the case when clock1 is stuck at 1 or 0.

Any error freezes the counters from counting. An application may then read out the counter values to help determine what caused the error.

Freezing Counters when Counter1 Reaches Zero:

The DCC module also allows the counters to be frozen when the counter1 reaches zero. This allows one of the clock sources for counter1 to be used as a reference for measuring one of the clock sources for counter0. The error conditions are the same as those where (counter0=0 and valid0=0) define the condition when the DCC counters are frozen. That is, an error is indicated if counter0 and valid0 become zero while counter1 is still non-zero. In this case, however, the application would typically set up the seed values such that the counter1 will become zero before counter0. Essentially the measurement period is defined by the seed value of the counter1. Note that this is also an error condition, and the interrupt service routine can use the measurement period and the actual cycles counted by counter1 to determine the frequency of the clock0 signal.

11.3 Clock Source Selection for Counter0 and Counter1

Refer to the device datasheet to identify the available options for selecting the clock sources for both counters of the DCC module. Some microcontrollers may include multiple instances of the DCC module. This will also be identified in the device datasheet.

The selection of the clock sources for counter0 and counter1 is done by a combination of the KEY, CNT0 CLKSRC, and CNT1 CLKSRC control fields of the CNT0CLKSRC and CNT1CLKSRC registers.

11.4 DCC Control Registers

This section describes the dual-clock comparator (DCC) module control and status registers. The registers support 8-bit, 16-bit or 32-bit writes and are aligned on a word (32-bit) boundary. [Table 11-1](#) shows address offsets from the module base address. The base address for the control registers is FFFF EC00h for DCC1 and FFFF F400h for DCC2.

Table 11-1. DCC Control Registers

Offset	Acronym	Register Description	Section
00h	DCCGCTRL	DCC Global Control Register	Section 11.4.1
04h	DCCREV	DCC Revision Id Register	Section 11.4.2
08h	DCCCNT0SEED	DCC Counter0 Seed Register	Section 11.4.3
0Ch	DCCVALID0SEED	DCC Valid0 Seed Register	Section 11.4.4
10h	DCCCNT1SEED	DCC Counter1 Seed Register	Section 11.4.5
14h	DCCSTAT	DCC Status Register	Section 11.4.6
18h	DCCCNT0	DCC Counter0 Value Register	Section 11.4.7
1Ch	DCCVALID0	DCC Valid0 Value Register	Section 11.4.8
20h	DCCCNT1	DCC Counter1 Value Register	Section 11.4.9
24h	DCCCNT1CLKSRC	DCC Counter1 Clock Source Selection Register	Section 11.4.10
28h	DCCCNT0CLKSRC	DCC Counter0 Clock Source Selection Register	Section 11.4.11

11.4.1 DCC Global Control Register (DCCGCTRL)

Figure 11-7 and Table 11-2 describe the DCC Global Control register.

Figure 11-7. DCC Global Control Register (DCCGCTRL) [offset = 00]

31															16																																																																
Reserved																																																																															
R-0																																																																															
15										12										11										8										7										4										3										0									
DONE INT ENA															SINGLE SHOT															ERR ENA															DCC ENA																																		
R/WP-5h															R/WP-5h															R/WP-5h															R/WP-5h																																		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

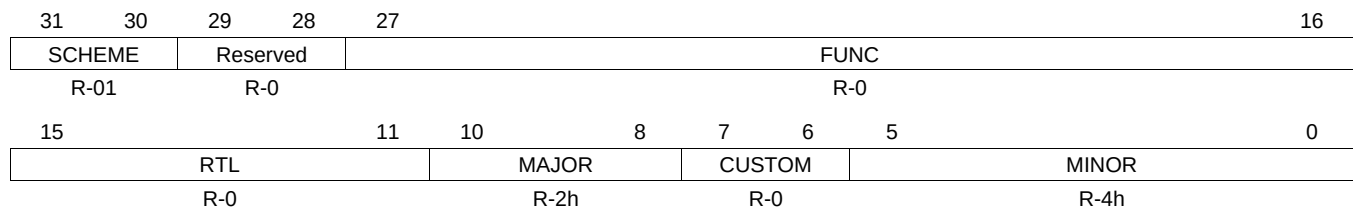
Table 11-2. DCC Global Control Register (DCCGCTRL) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-12	DONE INT ENA	5h No interrupt is generated when the DONE flag is set in the DCC Status (DCCSTAT) register. Others DONE interrupt is generated when the DONE flag is set in the DCC Status (DCCSTAT) register.	Done Interrupt Enable. Any operation mode read, privileged mode write:
11-8	SINGLE SHOT	Ah DCC stops counting when counter0 and valid0 both reach zero. Bh DCC stops counting when counter1 reaches zero. Others DCC counts continuously and only stops when an error occurs.	Single-Shot Mode Enable. Any operation mode read, privileged mode write:
7-4	ERR ENA	5h No interrupt is generated when the ERR flag is set in the DCC Status (DCCSTAT) register. Others ERROR interrupt is generated when the ERR flag is set in the DCC Status (DCCSTAT) register.	Error Interrupt Enable. Any operation mode read, privileged mode write:
3-0	DCC ENA	5h All DCC counters are stopped and error-checking is disabled. When an error occurs, the counters stop and this field is set to 5h automatically disabling the DCC counter in hardware. Others Read: Counters are enabled. Write: Load counters with their seed values and begin counting. It is recommended to write Ah to enable counters to protect against single-bit errors.	DCC Enable. Any operation mode read, privileged mode write:

11.4.2 DCC Revision Id Register (DCCREV)

Figure 11-8 and Table 11-3 describe the DCC Revision Id register.

Figure 11-8. DCC Revision Id Register (DCCREV) [offset = 4h]



LEGEND: R = Read only; -n = value after reset

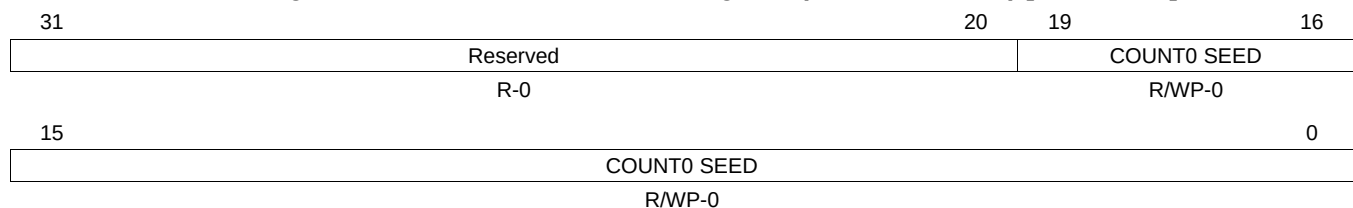
Table 11-3. DCC Revision Id Register (DCCREV) Field Descriptions

Bit	Field	Value	Description
31-30	SCHEME	01	Reads return 01, writes have no effect.
29-28	Reserved	0	Reads return 0. Writes have no effect.
27-16	FUNC	0	Functional release number. Reads return 0x000, writes have no effect.
15-11	RTL	0	Design release number. Reads return 0x00, writes have no effect.
10-8	MAJOR	2h	Major revision number. Reads return 0x2, writes have no effect.
7-6	CUSTOM	0	Custom version number. Reads return 0x0, writes have no effect.
5-0	MINOR	4h	Minor revision number. Reads return 0x4, writes have no effect.

11.4.3 DCC Counter0 Seed Register (DCCNT0SEED)

Figure 11-9 and Table 11-4 describe the DCC Counter0 Seed register.

Figure 11-9. DCC Counter0 Seed Register (DCCNT0SEED) [offset = 8h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 11-4. DCC Counter0 Seed Register (DCCNT0SEED) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	COUNT0 SEED		Seed value for DCC counter0. Reads in any operating mode return the current value of counter0. Writing in privileged mode only sets the current seed value for counter0.

NOTE: Seed for Counter0 must be non-zero

The DCC must only be enabled after programming a non-zero value in the COUNT0 SEED register.

11.4.4 DCC Valid0 Seed Register (DCCVALID0SEED)

Figure 11-10 and Table 11-5 describe the DCC Valid0 Seed register.

Figure 11-10. DCC Valid0 Seed Register (DCCVALID0SEED) [offset = Ch]

31	Reserved	16
	R-0	
15	VALID0 SEED	0
	R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 11-5. DCC Valid0 Seed Register (DCCVALID0SEED) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	VALID0 SEED		Seed value for DCC Valid0. This value defines the window within which the counter1 must reach 0. This window needs to be at least 4 cycles wide. Reads in any operating mode return the current value of seed for Valid0. Writing in privileged mode only sets the current seed value for Valid0. Writes in user mode are ignored.

NOTE: Seed for Valid0 must be at least 0x4

The DCC must only be enabled after programming a value greater than or equal to 0x4 in the VALID0 SEED register.

11.4.5 DCC Counter1 Seed Register (DCCCNT1SEED)

Figure 11-11 and Table 11-6 describe the DCC Counter1 Seed register.

Figure 11-11. DCC Counter1 Seed Register (DCCCNT1SEED) [offset = 10h]

31	Reserved	20	19	16
	R-0		COUNT1 SEED	
			R/WP-0	
15	COUNT1 SEED			0
	R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 11-6. DCC Counter1 Seed Register (DCCCNT0SEED) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	COUNT1 SEED		Seed value for DCC counter1. Reads in any operating mode return the current value of seed for counter1. Writing in privileged mode only sets the current seed value for counter1. Writes in user mode are ignored.

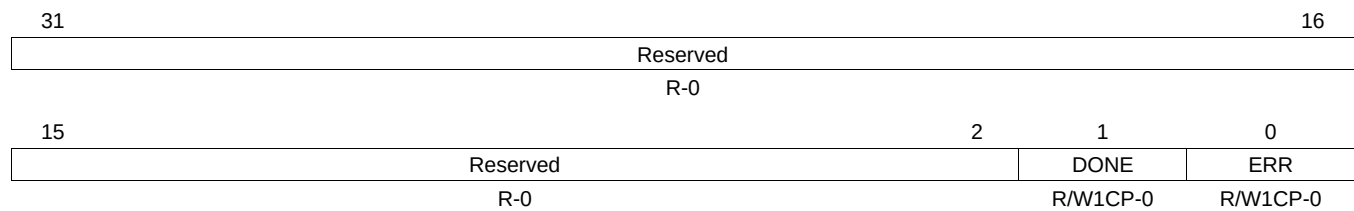
NOTE: Seed for Counter0 must be non-zero

The DCC must only be enabled after programming a non-zero value in the COUNT1 SEED register.

11.4.6 DCC Status Register (DCCSTAT)

Figure 11-7 and Table 11-2 describe the DCC Status register.

Figure 11-12. DCC Status Register (DCCSTAT) [offset = 14h]



LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 to clear in privilege mode only; -n = value after reset

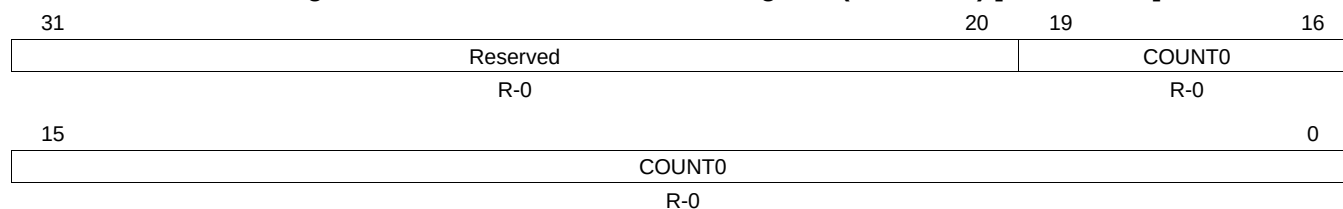
Table 11-7. DCC Status Register (DCCSTAT) Field Descriptions

Bit	Field	Value	Description
31-2	Reserved	0	Reads return 0. Writes have no effect.
1	DONE	0	Single-Shot Sequence Done flag. Indicates that a single-shot DCC sequence is done without any error. Read: Single-shot sequence is not done. Write: Writing 0 has no effect.
		1	Read: Single-shot sequence is done without any error. Write: Writing 1 in privileged mode clears the DONE flag.
0	ERR	0	Error flag. Indicates that a DCC error has occurred. Read: DCC error has not occurred. Write: Writing 0 has no effect.
		1	Read: An error has occurred. Write: Writing 1 in privileged mode clears the ERR flag.

11.4.7 DCC Counter0 Value Register (DCCCNT0)

Figure 11-13 and Table 11-8 describe the DCC Counter0 Value register.

Figure 11-13. DCC Counter0 Value Register (DCCCNT0) [offset = 18h]



LEGEND: R = Read only; -n = value after reset

Table 11-8. DCC Counter0 Value Register (DCCCNT0) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	COUNT0		Current value of DCC counter0. Reads in any operating mode return the current value of counter0. Writes have no effect.

NOTE: Reads may not return exact current value of counter

Reading the counter0 value while counting is enabled may not return the exact value of the counter0.

11.4.8 DCC Valid0 Value Register (DCCVALID0)

Figure 11-14 and Table 11-9 describe the DCC Valid0 Value register.

Figure 11-14. DCC Valid0 Value Register (DCCVALID0) [offset = 1Ch]

31	Reserved	16
	R-0	
15	VALID0	0
	R-0	

LEGEND: R = Read only; -n = value after reset

Table 11-9. DCC Valid0 Value Register (DCCVALID0) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	VALID0		Current value for DCC Valid0. Reads in any operating mode return the current value of Valid0. Writes have no effect.

NOTE: Reads may not return exact current value of Valid0

Reading the Valid0 value while counting is enabled may not return the exact value of the Valid0.

11.4.9 DCC Counter1 Value Register (DCCCNT1)

Figure 11-15 and Table 11-10 describe the DCC Counter1 Value register.

Figure 11-15. DCC Counter1 Value Register (DCCCNT1) [offset = 20h]

31	Reserved	20	19	16
	R-0		COUNT1	R/WP-0
15	COUNT1			0
	R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 11-10. DCC Counter1 Value Register (DCCCNT1) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-0	COUNT1		Current value for DCC counter1. Reads in any operating mode return the current value of counter1. Writes have no effect.

NOTE: Reads may not return exact current value of counter

Reading the counter1 value while counting is enabled may not return the exact value of the counter1.

11.4.10 DCC Counter1 Clock Source Selection Register (DCCCNT1CLKSRC)

Figure 11-15 and Table 11-10 describe the DCC Counter1 Clock Source Selection register.

Figure 11-16. DCC Counter1 Clock Source Selection Register (DCCCNT1CLKSRC) [offset = 24h]

31											16
Reserved											
R-0											
15	12	11							4	3	0
KEY		Reserved							CNT1 CLKSRC		
R/WP-5h		R-0							R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

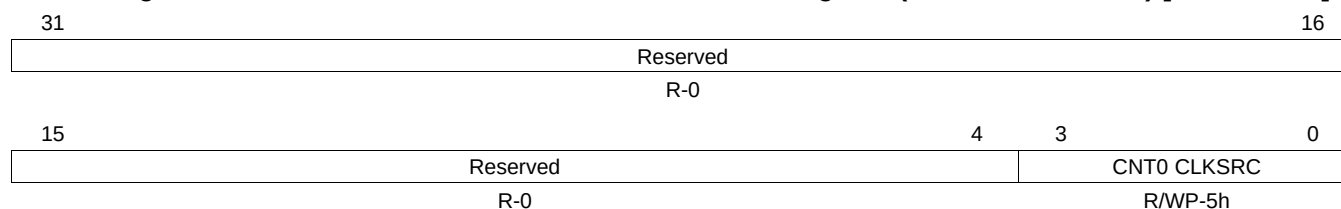
Table 11-11. DCC Counter1 Clock Source Selection Register (DCCNT1CLKSRC)
Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-12	KEY		Key to enable clock source selection for counter1. Reads in any operating mode return the current value of the key.
		Ah	Writes in privileged mode set the key value.
		Any other value	Writing Ah as the key enables the CNT1 CLKSRC field to define the clock source for counter1. Writing any other value as the key disables the clock source selection for counter1. In this case, the N2HET signal is used as the source for counter1. Refer to the device datasheet for available clock source options and the KEY required to enable these options for counter1.
11-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	CNT1 CLKSRC		Clock source for counter1 when KEY is programmed to Ah. Reads in any operating mode return the current value of CLKSRC. Writes in privileged mode select the clock source for counter1. Refer to the device datasheet for available clock source options and the KEY required to enable these options for counter1.

11.4.11 DCC Counter0 Clock Source Selection Register (DCCCNT0CLKSRC)

Figure 11-15 and Table 11-10 describe the DCC Counter0 Clock Source Selection register.

Figure 11-17. DCC Counter0 Clock Source Selection Register (DCCCNT0CLKSRC) [offset = 28h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

**Table 11-12. DCC Counter0 Clock Source Selection Register (DCCCNT0CLKSRC)
Field Descriptions**

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	CNT0 CLKSRC		Clock source for counter0 . Reads in any operating mode return the current value of CLKSRC. Writes in privileged mode select the clock source for counter0. Refer to the device datasheet for available clock source options for counter0.